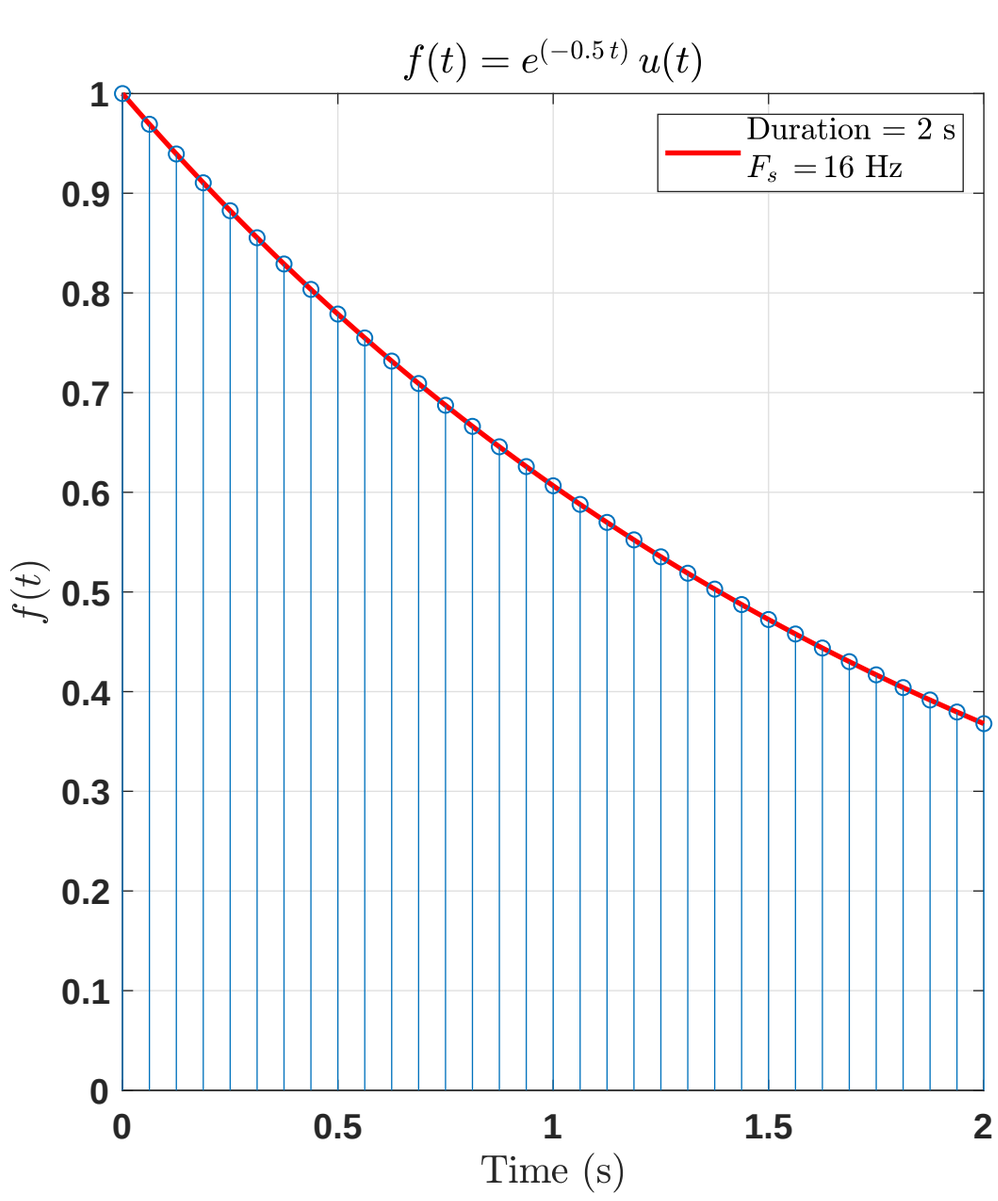
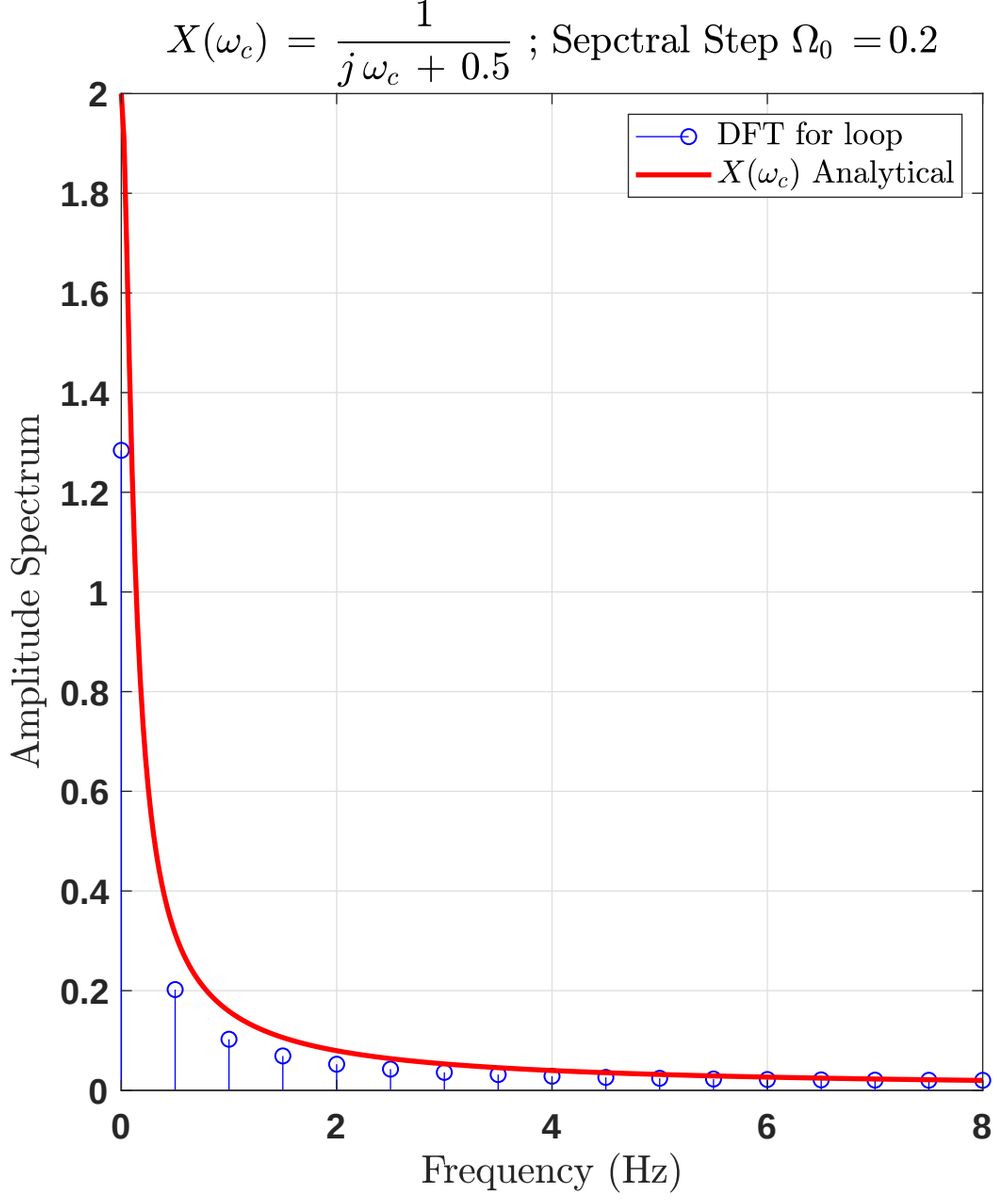

$f(t) = e^{(-0.5\,t)}\,u(t)$
Duration = 2 s
F_s = 16 Hz
$f(t)$
Time (s)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
0.5
1
1.5
2


$X(\omega_c) = \frac{1}{j\,\omega_c + 0.5}$; Sepctral Step Ω_0 = 0.2
DFT for loop
$X(\omega_c)$ Analytical
Amplitude Spectrum
Frequency (Hz)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0
2
4
6
8